Chenwan Halley Zhong

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EDUCATION

NORTHWESTERN UNIVERSITY Evanston, IL, U.S.
M.S. in Robotics *Sep 2025– Present*
BEIJING JIAOTONG UNIVERSITY Beijing, China
B.S.E. in Mechatronics *Sep 2020– Jul 2024*

SKILLS

Software: Python, C, C++, C#, MATLAB, PyTorch, OpenCV, HTML, JavaScript, CSS, Unity & Unreal Engine, Max/MSP
Robotics: ROS 2, MoveIt, Nav2, SLAM, Isaac Gym/Isaac Lab, MuJoCo, Gazebo, Inverse Kinematics, Contact Dynamics
Machine Learning & AI: Imitation Learning, LLM/VLM/VLA, Deep Learning, Reinforcement Learning, Diffusion Policy
Hardware: CAD (SolidWorks/Fusion 360), Blender, PCB(Altium/KiCad), Microcontrollers (STM32/Arduino/Raspberry Pi)
Languages: English (fluent), Mandarin Chinese (native), Cantonese, French, Spanish, German

EXPERIENCE

Stealth AI Startup - Robot Learning Engineer Intern *Jun 2026– Sep 2026*
- Deployed 10 Hz chunked diffusion policies on dual Flexiv arms with Wuji hands, interpolating to 1 kHz control
- Fine-tuned Robometer's categorical progress head on failed and rewound rollouts to detect task regressions
- Built a progress-gated state machine to sequence subtask policies using k-frame completion checks
- Engineered a leader-follower data pipeline with timestamp alignment and progress-based demo filtering

HKUST (GZ) Information Hub - Research Assistant *Oct 2024– May 2025*
- Developed an ESP32 embedded system in C to control a 10.2g rack-and-pinion module for a soft robotic interface
- Engineered a hardware-minimized PCB (3×3 cm) optimized for mobile, on-body robotic actuation
- Architected a GPT-4o & ElevenLabs pipeline to map generative theater scripts into synchronized robot performances
- Engineered a Python/MQTT pub-sub control layer for synchronized command dispatch across 6 robot nodes

Lenovo Research - Software Development Intern *Nov 2023– May 2024*
- Built a Unity (C#) 3D interaction pipeline integrating Leap Motion tracking and monocular camera input
- Retargeted hand poses to Blender-rigged skeletons using quaternion mapping, FK, and joint constraints
- Built 6 gesture recognizers using joint angles, fingertip distances, and hysteresis for 3D control

Nanyang Technological University (NTU) - Research Assistant *Jul 2023– Sep 2023*
- Analyzed commercial HVAC telemetry to extract critical power consumption features for cooling towers and chillers
- Developed a GNN architecture for thermal mapping, achieving an 8% error rate in sparse sensor environments
- Built high-fidelity digital twins in Modelica & EnergyPlus, validating 15% energy savings via optimized control

PROJECTS

GhostNote: Expressive Stiffness Control for a Robot Drummer *Apr 2026– Present*
- Built a 7-DoF OpenArm drummer that transfers human stiffness profiles onto animation-authored stroke trajectories
- Separated rebounding from damped strokes at AUC 1.00 via pre-impact co-contraction (8-ch EMG, MIDI-synced)
- Calibrated MuJoCo contact and grip dynamics against hardware, cutting wrist rebound error from 6.4 to 0.07 rad/s
- Optimized time-varying stiffness and grip via differential evolution: stick rebound fixed at 178 mm, now 66–293 mm
- Trained an MLP mapping target rebound to stiffness and grip profiles, achieving ~5 mm error on unseen targets

Vocal2Piano: End-to-End Autonomous Robotic Piano Accompaniment System *Jan 2026– Mar 2026*
- Built a piano accompanist with 2 belt-driven carriages and 26 solenoid fingers for live vocal accompaniment
- Built a humming-to-chord pipeline using YIN, HMM, and Viterbi decoding, achieving 87% synthetic note accuracy
- Mapped chord sequences to dual-carriage trajectories via Viterbi DP, enforcing timing and collision constraints
- Designed three 4-layer PCBs with protected 12 V input, dual stepper drivers, and 26 I²C-controlled solenoid channels

Bug Catcher: ROS 2 Vision-Based Dynamic Sorting (FER Panda) *Sep 2025– Dec 2025*
- Built a Python motion-planning client (Facade Pattern) wrapping ROS2 Service/Action calls into a MoveIt2 pipeline
- Calibrated a static TF tree via AprilTag averaging over 300 frames, reducing pose noise to <0.5 mm RMS
- Implemented an HSV blob-detection pipeline with a Persistent List tracker for real-time multi-target localization
- Optimized moving-target tracking via direct JointTrajectory injection at 20 Hz, bypassing MoveIt overhead